

Parent Guide — senior / module-06- moon-phases-tithi

IKS Curriculum

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Parent Guide — Moon Phases and Tithi (Senior)

What is your child learning?

The Moon's apparent change of shape across ~29.5 days is the synodic month. Indian astronomy divides this into 30 *tithis* (lunar days), grouped into a waxing fortnight (*śukla pakṣa*) and waning fortnight (*kṛṣṇa pakṣa*). This module connects observation, lunar geometry, calendar construction, and the panchanga framework.

Why this matters

This module is part of an Indian Knowledge Systems curriculum that introduces classical Indian frameworks alongside their modern-scientific parallels. Your child will learn to (a) name and use precise Sanskrit terms, (b) understand the observational basis for the classical framework, and (c) relate it to current scientific or scholarly understanding.

What you can do at home

This month, your child is keeping a 10-day moon diary. Please give them ~15 minutes each evening to observe the moon (a balcony or window is fine). Discussing what they see is part of the learning.

Conversation prompts

Try one of these over a meal or walk:

1. “What’s one Sanskrit word you learned this week? What does it mean?”
2. “What did you find surprising in this module?”
3. “Is there anything you’re not sure about? Let’s look it up together.”
4. “How would you explain this to your grandmother / grandfather?”

Questions parents often ask

Q: Is this religious instruction? A: No. The module presents Indian classical frameworks (cosmology, mathematics, astronomy, ecology) as historical knowledge systems with their own logic and method. The framing is scholarly and comparative, not devotional.

Q: Will my child be tested on Sanskrit? A: Sanskrit terms are introduced for precision, not for rote memorisation. The summative quiz asks students to *use* the terms — for example, “Which doṣa is dominant in the morning hours?” — rather than translate them.

Q: How does this connect to NCERT / board syllabus? A: This module supplements (does not replace) NCERT content. NEP 2020 §4.6 calls for the integration of Indian Knowledge Systems. The pack’s README.md lists the specific NEP principles each module advances.

Reading further (for parents)

See sources.md in this pack for the full source list.

10-Day Lesson Plan

Lesson Plan — Moon Phases and Tithi (Senior)

10-day arc, 45 min per session.

Day 1 — The Moon and Its Shapes

Objective: Observe and name the four basic phases (new, first quarter, full, last quarter).

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 2 — Two Fortnights — Pakṣa

Objective: Distinguish śukla pakṣa from kṛṣṇa pakṣa and name today’s.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 3 — Fifteen Tithis per Pakṣa

Objective: Recite the 15 tithi names (pratipadā, dvitīyā, ...) and explain why a tithi isn’t exactly a day.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 4 — Building a Tithi Calendar

Objective: Construct a tithi calendar for the current month using observation and computation.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 5 — Lunar Months and Festivals

Objective: Map the 12 lunar months (caitra, vaiśākha, ...) to major festivals.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 6 — Panchanga — Five Limbs

Objective: Identify the five components: tithi, vāra, nakṣatra, yoga, karaṇa.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 7 — Why Eclipses Happen

Objective: Explain solar/lunar eclipses using orbital plane geometry (5° tilt of Moon's orbit).

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 8 — Modern Lunar Science

Objective: Phases as Sun's illumination of Moon; mention libration, tides.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 9 — Cross-cultural Calendars

Objective: Compare with Hijri (Islamic), Chinese, and Hebrew lunar calendars.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 10 — Synthesis and Quiz

Objective: Final quiz and 10-day moon-diary share.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Pacing notes

- Senior sessions are 45 min. Adjust if your block is different.
- Days 5 and 9 are buffer days — use to reinforce weaker concepts or extend stronger ones.
- The summative project (see [assessments/project-brief.md](#)) runs in parallel from Day 6 onward.

Differentiation

- **For advanced students:** see [teacher-notes.md](#) for extension prompts.
- **For students needing more support:** simplify activities, do more pair-work, allow oral instead of written responses.

Homework

Homework — Moon Phases and Tithi (Senior)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — The Moon and Its Shapes** · Write 2 sentences about today's concept. (~10 min)
2. **Day 2 — Two Fortnights — Pakṣa** · Write 2 sentences about today's concept. (~10 min)
3. **Day 3 — Fifteen Tithis per Pakṣa** · Write 2 sentences about today's concept. (~10 min)
4. **Day 4 — Building a Tithi Calendar** · Write 2 sentences about today's concept. (~10 min)
5. **Day 5 — Lunar Months and Festivals** · Write 2 sentences about today's concept. (~10 min)
6. **Day 6 — Panchanga — Five Limbs** · Write 2 sentences about today's concept. (~10 min)
7. **Day 7 — Why Eclipses Happen** · Write 2 sentences about today's concept. (~10 min)
8. **Day 8 — Modern Lunar Science** · Write 2 sentences about today's concept. (~10 min)
9. **Day 9 — Cross-cultural Calendars** · Write 2 sentences about today's concept. (~10 min)
10. **Day 10 — Synthesis and Quiz** · Write 2 sentences about today's concept. (~10 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.